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A detector for measuring the CO2 emissions in the atmosphere

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Abstract

Pollution has a major role in damaging our earth. The rapid industrialization, fast urbanization, rapid growth in population, drastic increase in vehicles on roads and other activities of human beings have disturbed the balance of natural atmosphere. It changes the quality of climate and climate change is brought about by the accumulation of greenhouse gases in the atmosphere. One of the greatest environmental problems facing the world today is global warming caused by emission of greenhouse gases. Cabon dioxide, which is an important constituent of environment is causing a warming effect on the earth's surface. To save our environment, monitoring and controlling these changes is a big challenge. In terms of long-range control of CO2 emission at their source is a more desirable and effective method through preventive and control technologies. The main objective of this project is to implement a CO2 detector to measure the amount of CO2 emission in the atmosphere with a LED that act as a signal when the CO2 emissions measured by the MG-135 exceed a 1000 ppt.

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1 Chapter 1. Introduction

1.1 Description of the Project

Carbon dioxide emissions, largely by-products of energy production and use, account for the largest share of greenhouse gases, which are associated with global warming. The atmospheric concentration of carbon dioxide, the major greenhouse gas directly affected by human activities, has increased by about 30% since the beginning of the pre- industrial era around 1750 because of the combustion of fossil fuels and changes in land- use practices [2].

It is the principal anthropogenic greenhouse gas that affects the Earth's radiative balance. This is leading to an increase in the earth's surface temperature and related effects on climate, sea level rise, and world agriculture [1]. Globally, CO2 emissions are growing at an alarming rate, primarily due to the burning of fossil fuels for energy production and transportation. Without significant reductions in CO2 emissions, the impacts of climate change will continue to worsen.

So, to stay informed about the percentage of emissions in the atmosphere and to take action to reduce the carbon footprint, our project is a detector system that continuously monitors CO2 levels in the environment and provides real-time data to help making decisions to reduce carbon footprint. By increasing awareness and promoting sustainable practices, we can work towards mitigating the effects of climate change and protecting our planet for future generations.

1.1.1 Why is carbon dioxide harmful to the environment?

1.1.1.1 Health impact

Symptoms of mild CO2 exposure may include headache and drowsiness. At higher levels, rapid breathing, confusion, increased cardiac output, elevated blood pressure, and increased. arrhythmias may occur. Breathing oxygen depleted air caused by extreme CO2 concentrations can lead to death by suffocation.[3]

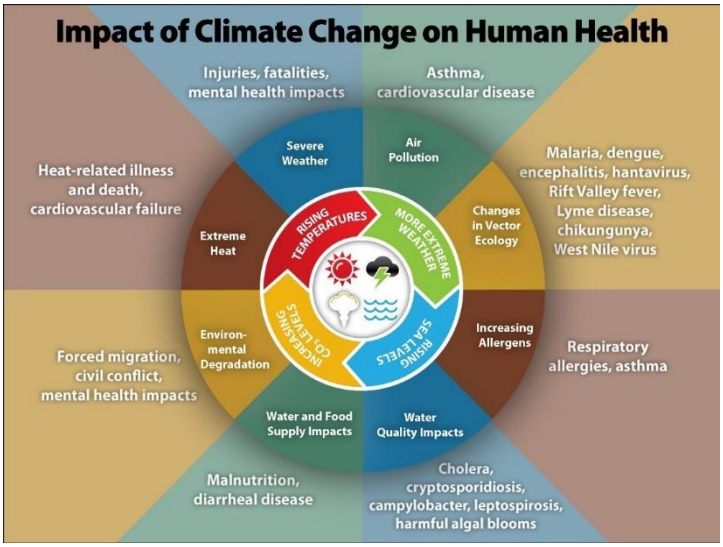


Figure 1: Impact of climate change on human healthcare.[18]

1.1.1.2 Agriculture impact

According to studies, rising carbon dioxide levels will have negative impacts on crop yields in some cases. The negative impacts include reduced yields due to higher temperatures and drier conditions. For example, maize yields are expected to fall by 15% in areas that rely on irrigation and by 8% in areas that rely on rain. Rainfed crops in arid climates, such as southern Africa and India, are expected to see a decrease in yield of 29% without the CO2 boost. This is because these crops are already water-stressed, and increased CO2 levels will not be enough to offset the negative impacts of reduced water availability.[4]

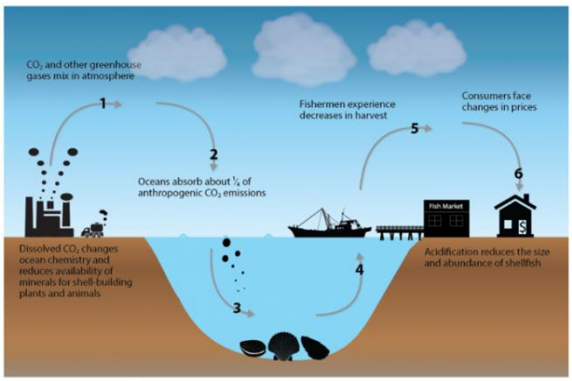


Figure 2: Ocean Acidification Impact for Shellfish.[4]

1.1.1.3 Weather pattern impact

According to studies, CO2 emissions trap the sun's heat which leads to global warming and disrupts the usual balance of nature. This warming disrupts weather patterns in several ways including causing more extreme weather events and making some areas more prone to drought.[5]

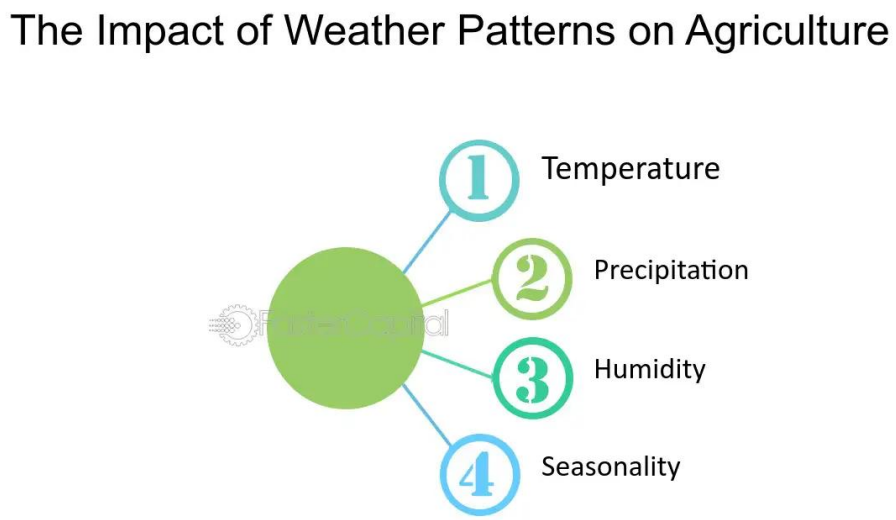


Figure 3: The warming disrupts weather patterns in several ways.[17]

1.1.1.4 More health risks

Climate change is the single biggest health threat facing humanity. Climate impacts are already harming health, through air pollution, disease, extreme weather events, forced displacement, pressures on mental health, and increased hunger and poor nutrition in places where people cannot grow or find sufficient food.

Every year, environmental factors take the lives of around 13 million people. Changing weather patterns are expanding diseases, and extreme weather events increase deaths and make it difficult for health care systems to keep up.

1.2 Overview of similar project

Project 1:

Figure 4: Carbon monoxide detector with LED display temperature and humidity monitoring.[6]

Project 2:

Figure 5: Uses lasers to monitor carbon dioxide.[19]

Table 1: Overview of similar project.[6][19]

No.	Criteria	Project 1	Project 2
1	**Design**	**Advantages:** -The 6-in-1 carbon monoxide detector just monitors CO, but accurately measures CO2, TVOC, HCHO, temperature and humidity level. -Equipped with high precision electrochemical sensor.	**Advantages:** -Drive to improve greenhouse gas-monitoring abilities at CO2 storage sites. -A multipoint laser-based system to provide around-the-clock monitoring to detect potential leaks. -The GreenLITE system runs autonomously and can send text message updates. **Disadvantages:** -It can take as long as a month for data to be collected and analyzed.
2	**Ease of use**	**Advantages:** -Carbon monoxide detector with wide measuring ranges, high accuracy, small size, easy operation. Suitable for indoor and outdoor, home, office, school, camping, picnic, boat, plane, car.	**Disadvantages:** -Capture technologies take CO2 from exhaust or reformed gases or stationary sources using pre-combustion, post-combustion or oxy-combustion techniques. -Sequestration technologies deposit captured CO2 in geological formations.
3	**Cost**	**100-150 SAR**	**15000-30000 SAR**

1.3 Objectives

1. Design a detector system for carbon dioxide emissions in the atmosphere.
2. Raise awareness of human activities contributing to climate change and the resulting health impacts.
3. Design an affordable CO2 emission detector system in low-resource environments to help people own it.

1.4 Problem definition

The CO2 detector is one of the most significant items in monitoring and reducing greenhouse gas emissions since it may protect both human health and the environment from the harmful effects of climate change, such as extreme weather events, rising sea levels, and

This underscores the importance of implementing a cost-effective CO2 detection system in low-resource environments where access to such technology may be limited. By doing so, communities can better understand their impact on the environment and take steps to reduce emissions.

The United Nations Framework Convention on Climate Change (UNFCCC) states that "stabilization of greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system is a common but differentiated responsibility and requires deep cuts in global emissions of greenhouse gases."[7]

If nothing changes, global atmospheric CO2-concentrations could reach 550 parts per million (ppm) in 2050.[8]

Therefore, the problem addressed in this project may be summarized as: to implement a CO2 detector system that can monitor CO2 emissions with a signal that turn when the emission reach above 1000 ppt, to have students apply what they have learned in academic courses practically and have hands-on experience with real physical models.

The project aims to reach a desired state accomplishing its set objectives. Thus, a table

showing the present state this project aims to improve on after completing it.

Table 2: present and desire state of the project.

Desired State	Present State
Students have no inexpensive practical tools to apply the knowledge achieved during academic studies.	To have an inexpensive CO2 detector to apply what students learn in academic courses on real physical models and have firsthand practical experience with such concepts.
people lack knowledge about the amount of co2 emissions around them	To have a device that can provide real-time information on CO2 levels in the surrounding environment

1.4.1 Statistics of CO2 emissions in Saudi Arabia

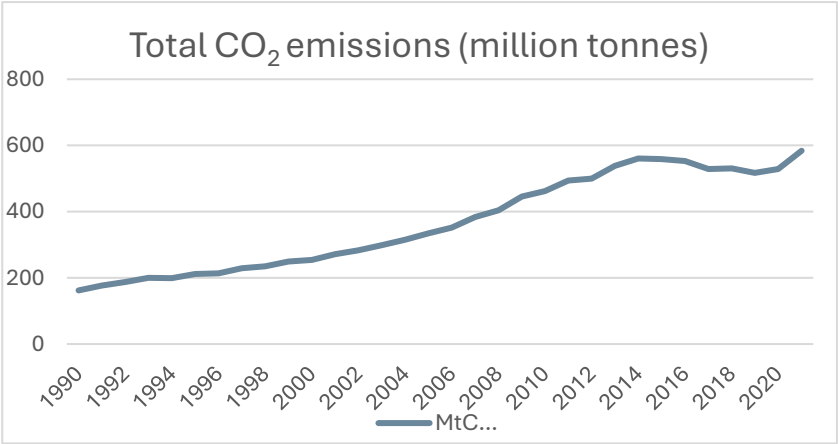


Figure 6: Carbon dioxide emissions in Saudi Arabia increased steadily over the first years.[9]

Saudi Arabia is one of the top emitters of CO2 globally due to its heavy reliance on oil and gas production. Implementing a CO2 detector system in the country could help track and reduce these emissions effectively.

According to the data, Saudi Arabia's CO2 emissions have been steadily increasing over the years, making it crucial to address this issue through practical solutions like the proposed detector system.

Numbers have shown that, in 2021, CO2 emissions had reached 584 million metric tons. This increase wouldn't stop without proactive measures to monitor and control emissions. [9]

2 Chapter 2. Method and Materials

In order to effectively implement the CO2 detector system, it is important to outline the methods and materials that will be used for tracking emissions.

This chapter will detail the specific materials and methods necessary for accurately measuring CO2 levels.

2.1 Materials

2.1.1 Arduino

2.1.1.1 Description of Arduino

A microcontroller is a type of computer in the form of an integrated circuit (IC) or "chip". Microcontrollers are often used in portable devices which run on batteries, e.g. digital cameras. So they are often low-powered with a small current consumption. Microcontrollers are programmable, which means that they can be customized to perform specific tasks. Some of the commonly used programming languages include C, C++, and assembly language. [10]

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino consists of both a physical programmable circuit board which is type of microcontroller, and a IDE (Integrated Development Environment), used to write and upload computer code to the physical board.[11]

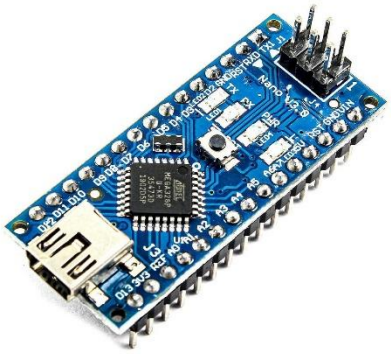


Figure 7 : Arduino Nano.[11]

2.1.1.2 What is Arduino used for?

Arduino serves as the central processing unit for CO2 detection circuit. It's responsible for reading sensor data, displaying the values, and potentially controlling outputs or data logging based on the program. [11]

2.1.1.3 Why Arduino is the best microcontroller?

Arduino is ideally suited for projects that bridge the gap between the physical world and the digital world. In essence, Arduino boards are microcontroller development platforms. This means they can be programmed to take input from the real world through sensors and control outputs such as lights, or even send data to OLEDs.[11]

2.1.2 Resistor

2.1.2.1 Description of Resistor

Resistor is defined as a passive electrical component with two terminals. In almost all electrical networks and electronic circuits they can be found and it’s measured in ohms. [12]

Figure 8 : A picture of a resistor.[12]

2.1.2.2 What is resistor used for?

Resistors limit the current flow in a circuit. They are crucial for protecting the components, especially the OLED display and potentially the CO2 sensor, from damage caused by excessive current.

2.1.2.3 Why are resistor chosen?

CO2 sensors and OLED displays often have specific voltage requirements. Resistors ensure the correct amount of current reaches these components by regulating the flow.

2.1.3 Wires

2.1.3.1 Description of Wires

wires are simply wires that have connector pins at each end, allowing them to be used to connect two points to each other without soldering. [13]

Figure 9: A set of jumper wires.[16]

2.1.3.2 What are wires used for?

Wires act as the conductive pathways that connect all the components in the circuit, allowing them to communicate and exchange data.

2.1.3.3 Why are wires chosen?

Without wires, the components would be isolated units and wouldn't be able to function as a complete circuit. They carry the electrical signals between the Arduino, CO2 sensor, and OLED display.

2.1.4 The OLED screen.

2.1.4.1 Description of the OLED screen

The Organic Light-Emitting Diode, or simply OLED, is the technology used for the screen in the Arduino Sensor Kit. The OLED uses an organic, carbon-based material for emitting light.

When electricity is applied to this material, it emits light. The Grove OLED display 0.96" module's behavior can be controlled by using the Adafruit_SSD1306.h library.[11]

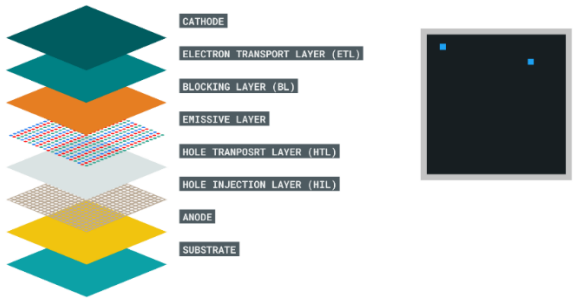


Figure 10: Organic Light-Emitting diode.[16]

2.1.4.2 What is OLED used for?

An OLED (Organic Light-Emitting Diode) display acts as the visual output device in the circuit. It allows to display the CO2 concentration readings obtained from the sensor in a clear and user-friendly format.

2.1.4.3 Why is it important for the project?

The OLED display provides a real-time visualization of the CO2 levels detected by the sensor. This allows to easily monitor and interpret the CO2 concentration data.

2.1.5 Breadboard

2.1.5.1 Description of Breadboard

A Breadboard is simply a board for prototyping or building circuits on. It allows to place components and connections on the board to make circuits without soldering.[11]

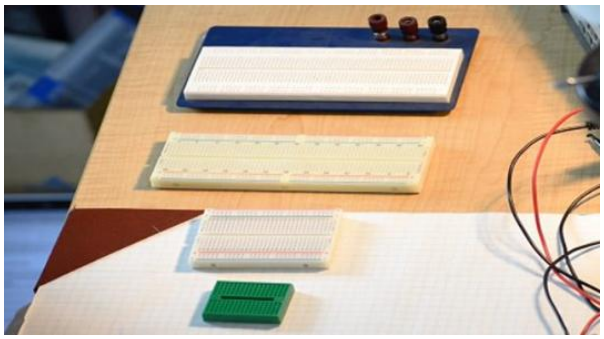


Figure 11: Different breadboard sizes.[16]

2.1.5.2 What is breadboard used for?

A breadboard is a temporary prototyping platform that allows you to connect electronic components without soldering. It has rows of holes connected electrically, making it easy to insert component legs and wires to create circuits. [11]

2.1.5.3 Why breadboard chosen for the project?

The breadboard provides a convenient and reusable platform to build the CO2 detection circuit. Provide easy experiments with different configurations and troubleshoot any issues before finalizing the design.

2.1.6 MG- 135 Sensor

2.1.6.1 Description of MG-135 Sensor

An MQ135 air quality sensor is one type of MQ gas sensor used to detect, measure, and monitor a wide range of gases present in air like ammonia, alcohol, benzene, smoke, carbon dioxide, etc. It operates at a 5V supply with 150mA consumption. Preheating of 20 seconds is required before the operation, to obtain the accurate output.[15]

Figure 12: MQ135 air quality sensor.[15]

2.1.6.2 What is it used for?

In this project, MG-135 is used to monitor and measure CO2 emissions in the atmosphere and provide data to display on a digital screen for real-time tracking.[15]

2.1.6.3 Why is it chosen?

The MQ-135 sensor's affordability and ease of use make it suitable. Additionally, its compact size and low power consumption make it a popular choice for this project.[15]

2.1.7 LEDs

2.1.7.1 Description of LED

An LED is a semiconductor device that emits light when an electric current passes through it. [16]

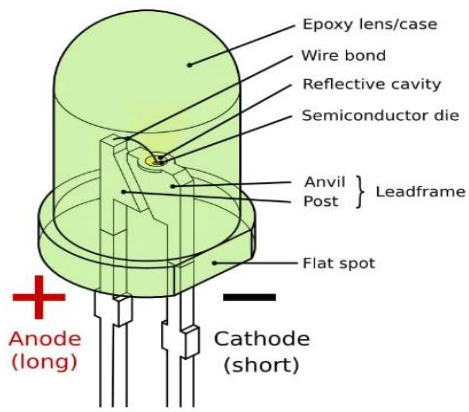


Figure 13: The component of the LED.[16]

2.1.7.2 Why LED chosen for this project?

LEDs are known for their low power consumption, reliability, and long lifespan, making them a suitable choice for your project.

2.1.7.3 What is it used for?

LED act as an indicate when the CO2 emissions measured by the MG-135 sensor exceed a certain threshold, specifically 1000 ppt (parts per thousand).

2.1.8 PLA

2.1.8.1 Description of PLA

PLA is a thermoplastic polyester. PLA has become a popular material due to it being economically produced from renewable resources and in 2010, PLA had the second highest

consumption volume of any bioplastic of the world and considered the most widely used plastic filament material in 3D printing and has a leading biomaterial for numerous applications in medicine as well as in industry replacing conventional petrochemical-based polymers. PLA was synthesized in 1932 by Carothers (at DuPont) [16].

Figure 14: Picture or regular PLA.[16]

2.1.8.2 What is PLA used for?

Used as a 3D printing feedstock for desktop fused filament fabrication 3D printers. PLA is popular for 3D printing as it can easily be.

2.2 Software

2.2.1 Arduino IDE

2.2.1.1 Description of Arduino IDE

The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino hardware to upload programs and communicate with them. [11]

Programs written using Arduino Software (IDE) are called sketches. These sketches are written in the text editor and are saved with the file extension. ino. [11]

Figure 15: Arduino software (IDE).[11]

2.2.1.2 What is Arduino IDE used for?

Coding in Arduino IDE will specify how to Read data from the CO2 sensor, Interpret the sensor readings and convert them into meaningful CO2 concentration values.[14]

Once writing the code, Arduino IDE compiles it into a format that the Arduino board can understand and execute. This compilation process ensures the code is compatible with the board's hardware.[14]

2.2.1.3 Why was Arduino IDE chosen?

Arduino IDE is important to write the code that instructs the Arduino board how to function. Arduino IDE serves a vital role in the software development environment. It's a free, open-source software application specifically designed to write code for Arduino boards.[14]

2.2.2 Solid Works

2.2.2.1 Description of Solid Works

Solid Works is used to develop a robust and fully functioning mechatronics system from beginning to end. At the initial stage, the software is used for planning, visual ideation, modelling, prototype, and project management. Finally, you can manage things like device management, analytics, data automation, and cloud services with the help of Solid Works [16].

The SOLIDWORKS software is efficiently used by mechanical, electrical, or ECE engineers to invent a new connected design. The program suite aims to keep engineers in sync, communicate and respond to design needs or any changes [16].

2.2.2.2 What are SolidWorks used for?

SolidWorks is used to create 3D CAD models of the case and visualize it before the manufacturing process to verify the design and its aspects.[16]

2.2.2.3 Why Solid Works was chosen.

Flexible and expanded functionality of the SOLIDWORKS Manage allows easy and detailed planning of each project stage or process progress, including distribution of defined tasks to desirable departments or employees [16].

2.2.3 Ultimaker Cura

2.2.3.1 Description of Cura

For 3D printers, Cura is an open-source slicing program. The LGPLv3 license applies to Cura. The LGPLv3 license replaced the open source Affero General Public License version 3 that Cura was initially distributed under on September 28, 2017. This modification made it possible to integrate third-party CAD applications more thoroughly. Although it may be used with other printers as well, it is the primary 3D printing software for Ultimaker printers [16].

Figure 16: The application ultimaker cura defaut image.[16]

2.2.3.2 what is Ultimaker Cura used for

Cura slices 3D models. In other words, it translates your 3D printing file (containing a 3D model you drew in CAD software) into G-code. G-code is the language your 3D printer understands, with the sliced file containing print instructions for your 3D printer [16].

2.2.3.3 What makes Ultimaker Cura the most advanced 3D printer software

- **Slicing features**

At the heart of Ultimaker Cura is its powerful, open-source slicing engine, built through years of expert in-house development and user contributions .[16]

- **Integrated workflow**

If you own a 3D printer, software matters. Get the most from your printer with software designed to suit your workflow. [16]

- **Easy to use**

Manufacturing doesn't need to be complicated. We design our software so anyone can use it – both experienced or rookie 3D printer users.[16]

2.3 Manufacturing methods

2.3.1 3D Printer (Ender 3 V2)

2.3.1.1 Description of 3D printer

3D printing, or additive manufacturing, is the process of creating a three-dimensional object from a computer-aided design (CAD) model. This technology increases designer productivity, improves design quality, and improves communication through documentation. 3D printing can involve various processes where material is deposited, joined, or solidified under computer control to create a three-dimensional object layer by layer. [16]

Figure 17: Picture of a FDM printer.[16]

2.3.1.2 What is 3D printer used for?

The Ender 3 3D printer is a computer-aided design process that can create complex shapes or geometries that are impossible to construct by hand.[16]

It can produce hollow parts or parts with internal truss structures to reduce weight. Fused deposition modeling (FDM) is the most common 3D printing process in use as of 2020. 3D printing involves creating a three-dimensional object.[16]

from a CAD model, which is computer-aided design. This process increases designer productivity, improves design quality, improves communication, and creates a database for manufacturing. 3D printing can also involve depositing, joining, or solidifying material under computer control to create a three-dimensional object layer by layer.[16]

2.3.1.3 Why 3D printer is chosen.

Ease and adjustability of designs. The possibility of designing all models and complex designs. Save time and effort because production cycles are short. Printing everything that cannot be printed by traditional methods of parts, whether large in size or prominent parts and overlapping with each other.[16]

2.4 Methods

2.4.1 project Specification

2.4.1.1 Circuit specification

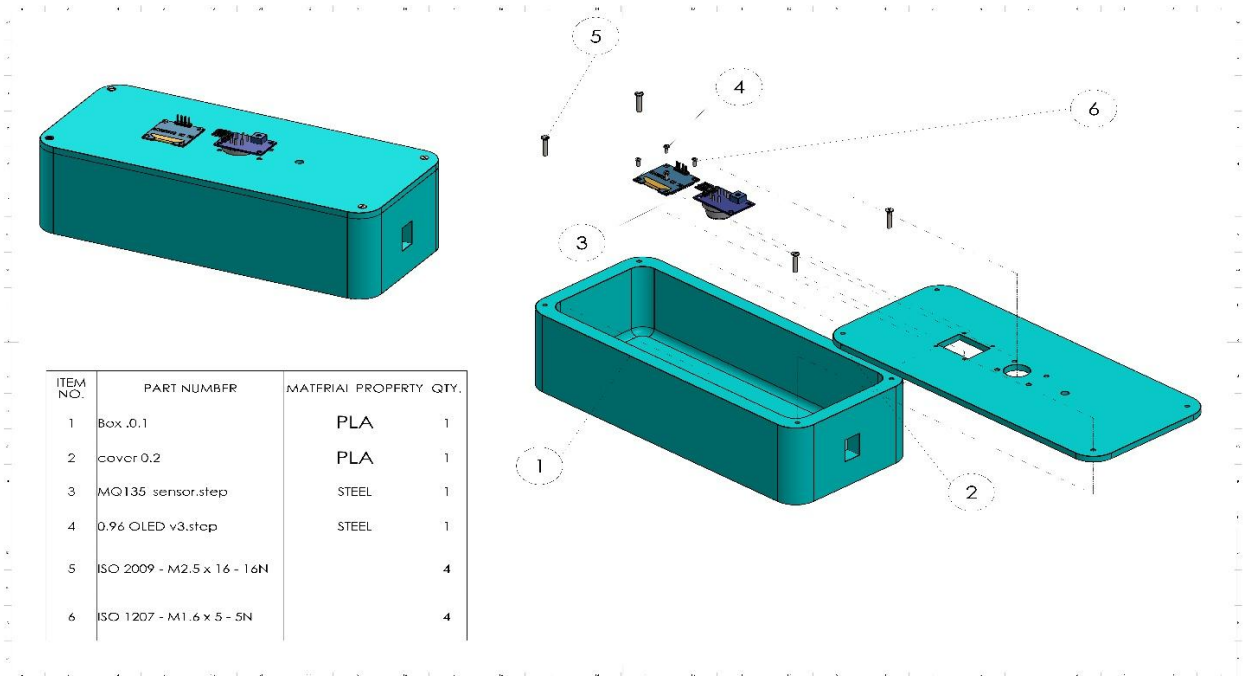
The model was designed using an Arduino Uno microcontroller, MG-135 sensor, LED, and OLED display screen module for Arduino. The design specification of the proposed system is described in Table 3.

Table 3: Circuit Specification.

S/N	Component	Quantity
1	Arduino Uno	1
2	MQ-135 Sensor	1
3	Bread Board	1
4	120 Ohm Resistor	1
5	OLED Display	1
6	Connecting Wires	Many
7	9V Battery	1
8	Red Led	1
9	Battery Switch	1

2.4.1.2 Case specification

SolidWorks, a 3D design software, was used to create a case for the system that would securely hold all the components in place and protect them. The design includes proper openings for the OLED, LED, the sensor, and the battery switch.



2.4.2 Circuit of measuring CO2 emissions

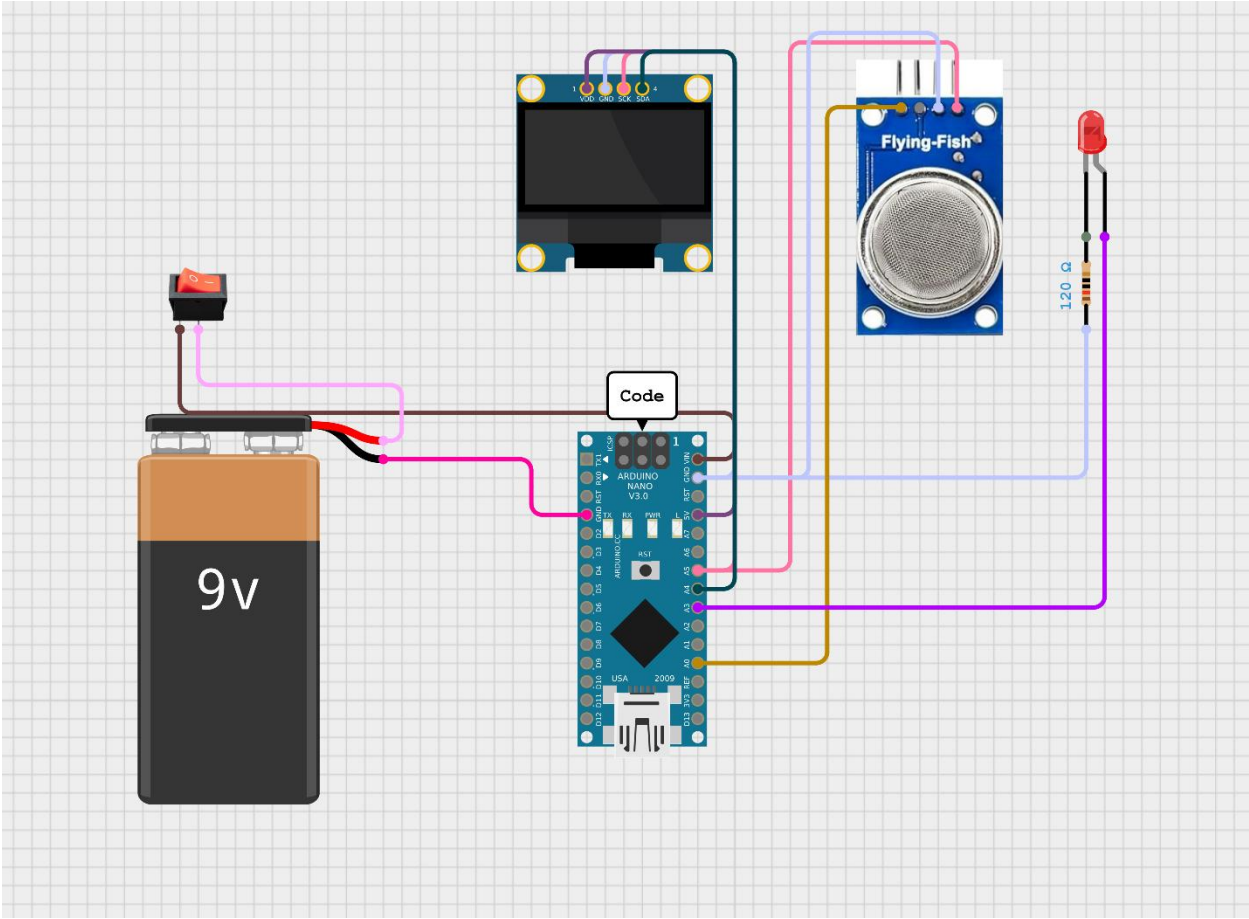


Figure 18: Circuit of measuring CO2 emissions.

The circuit includes connections between the battery button, MG-135 sensor, OLED screen, and LED. These components work together to accurately measure and display CO2 emissions levels. The circuit design ensures efficient communication between the components, allowing for real-time monitoring of CO2 levels. This system provides a visual warning when emissions reach a critical level, enhancing safety and awareness.

2.4.3 Project flow chart

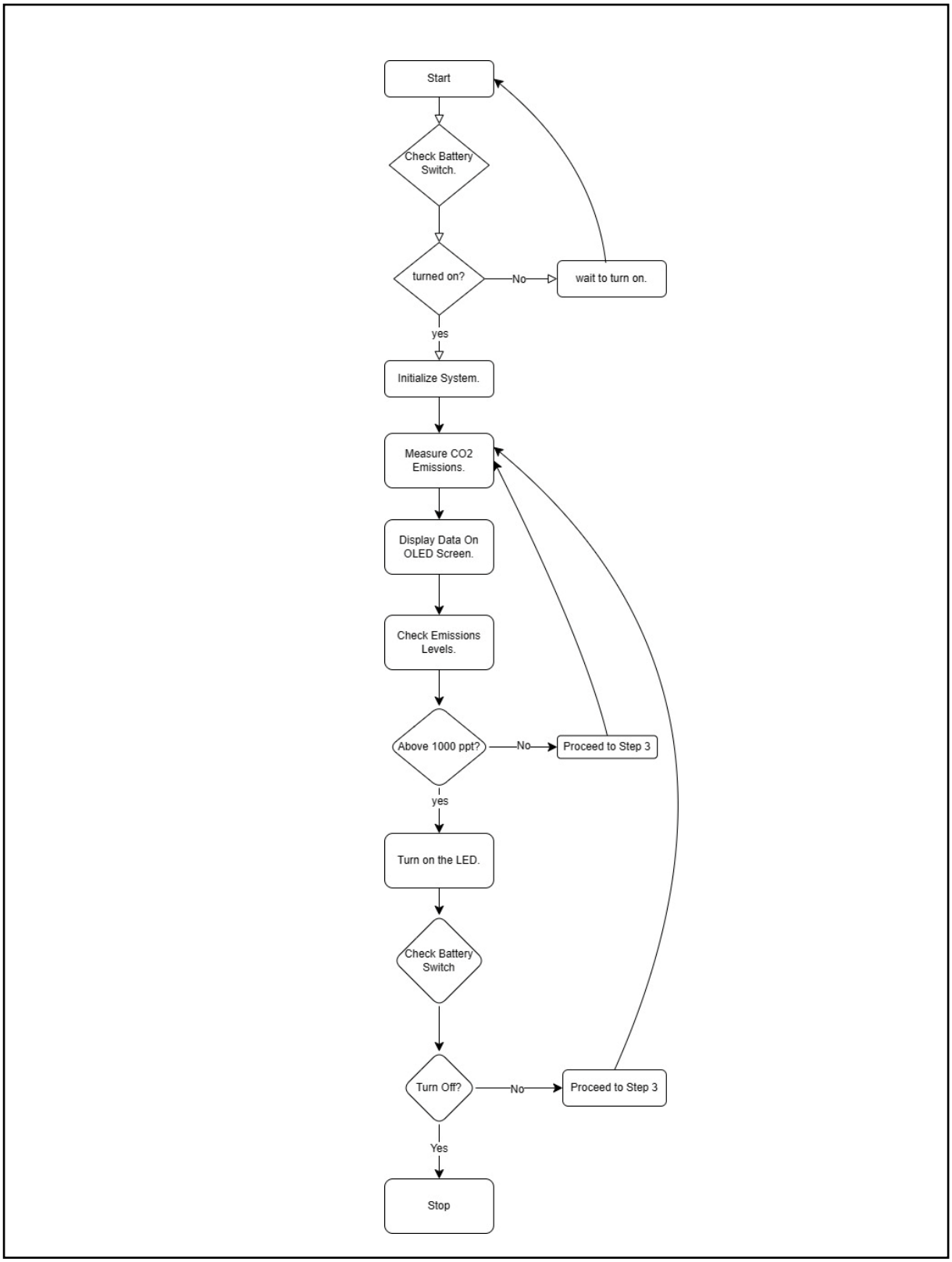


Figure 19: Flow Chart of the project.

As described in figure 18, the battery switch turns on, then the MG-135 sensor, OLED screen, and the LED are initializing. The MG-135 Sensor starts to read CO2 emissions, then displays the data on the OLED screen. The LED illuminates as a signal when the emissions level reaches 1000 parts per million. Until the battery switch is turned off, the system will keep tracking the CO2 emissions.

3 Chapter 3. Results and Discussion

3.1 Observations during the prototyping stage

Through the process of prototyping, a system's concept is tested to see if its components and coding perform as expected.

3.1.1 Testing the circuit

This step is crucial in determining the system's reliability and effectiveness in monitoring CO2 emissions levels. The circuit was designed using a sensor, OLED, LED, and the microcontroller Arduino Nano. The sensor detects CO2 levels, which are then displayed on the OLED screen and indicated by the LED. The Arduino Nano processes this data and ensures timely alerts for any dangerous levels of CO2.

Through this step, the functionality of the circuit can be verified, ensuring that it accurately measures and displays CO2 levels for effective monitoring.

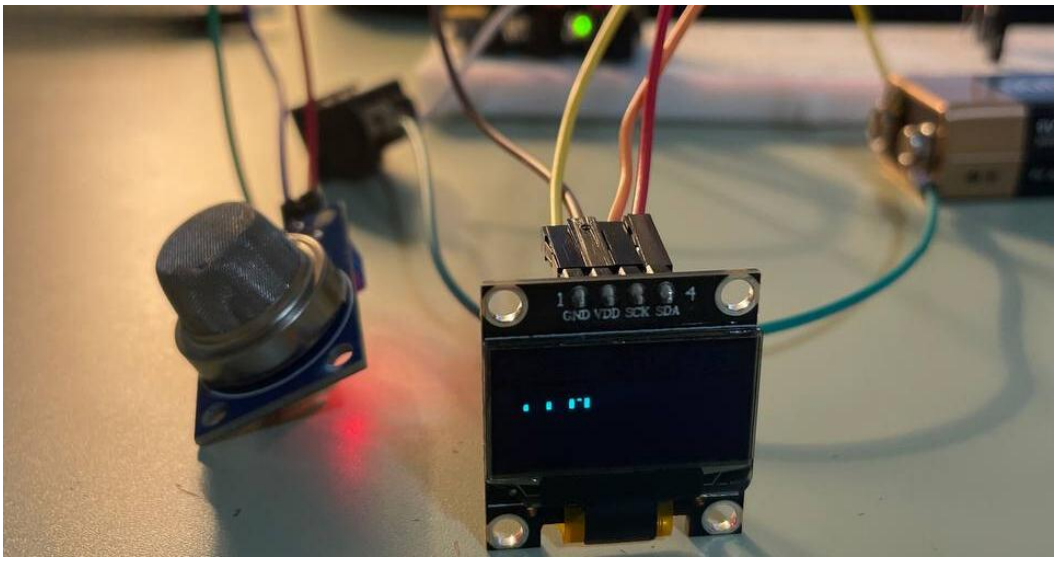


Figure 20: testing the circuit.

3.1.2 Testing the code and its libraries

Once the code is uploaded to the Arduino Nano, it is important to test its functionality and ensure that the libraries used are compatible with the components. This testing phase is crucial in identifying any potential errors or bugs that may affect the accuracy of CO2 level monitoring.

The code utilized in this project consists of declaring the following libraries: <MQ135.h>, <Adafruit_GFX.h>, <Wire.h>, and <Adafruit_SSD1306.h>. Next, specify the pins that the Arduino Nano will use to communicate with all of its components. Finally, the void setup for the code itself, which displays the data and determines whether the emissions were above the maximum, 1000ppt, amount.

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <MQ135.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET    -1
#define SENSOR_PIN    A0
#define LED_PIN       A3

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);
MQ135 gasSensor = MQ135(SENSOR_PIN);

void setup() {
  Serial.begin(9600);
  pinMode(LED_PIN, OUTPUT);
  if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println(F("SSD1306 allocation failed"));
    for(;;);
  }
  display.clearDisplay();
  display.display();
}

void loop() {
  int val = analogRead(SENSOR_PIN);
  float ppm = gasSensor.getPPM();
  Serial.print("CO2 PPM: ");
  Serial.println(ppm);

  display.clearDisplay();
  display.setTextSize(2);
  display.setTextColor(WHITE);
  display.setCursor(0, 0);
  display.print("CO2: ");
  display.print(ppm);
```

```
  display.print(" PPM");
  display.display();

  if(ppm > 1000) {
    digitalWrite(LED_PIN, HIGH);
  } else {
    digitalWrite(LED_PIN, LOW);
  }

  delay(2000);
}
```

Figure 22: the source code of the project.

3.1.2.1 Wire.h library

The Wire.h library in Arduino allows to communicate with I2C devices. It's a common protocol primarily used for reading from and sending data to external I2C components. **[11]** In this project, Wire.h library used to communicate with the CO2 sensor and display the readings on the screen. Additionally, a conditional statement used to control an LED based on the CO2 levels detected by the sensor.

3.1.2.2 MQ135.h library

The MQ135.h library allows easy interfacing with the MQ135 air quality sensor in Arduino projects. It calculates CO2 concentrations and provides a straightforward way to read data from the sensor. [11]

3.1.2.3 Adafruit_GFX.h library

The Adafruit_GFX.h library is a fundamental graphics library for Arduino that provides a set of functions for drawing shapes, text, and images on displays. It simplifies creating graphical user interfaces (GUIs) and enables efficient rendering on various display types.

3.1.2.4 Adafruit_SSD1306.h library

The Adafruit_SSD1306.h library is a graphics library for Arduino that simplifies interfacing with SSD1306-based OLED displays. It provides functions for drawing shapes, text, and images on these displays, making it easier to create custom user interfaces and visual elements.

3.1.3 Testing the case

Testing the case involves ensuring that the components fit properly and that the connections are secure. It is important to conduct thorough testing to verify the functionality and durability of the overall project.

To ensure that the components fit correctly, a CAD model was used to insert the models into the case and copy its dimensions for precise measurements.

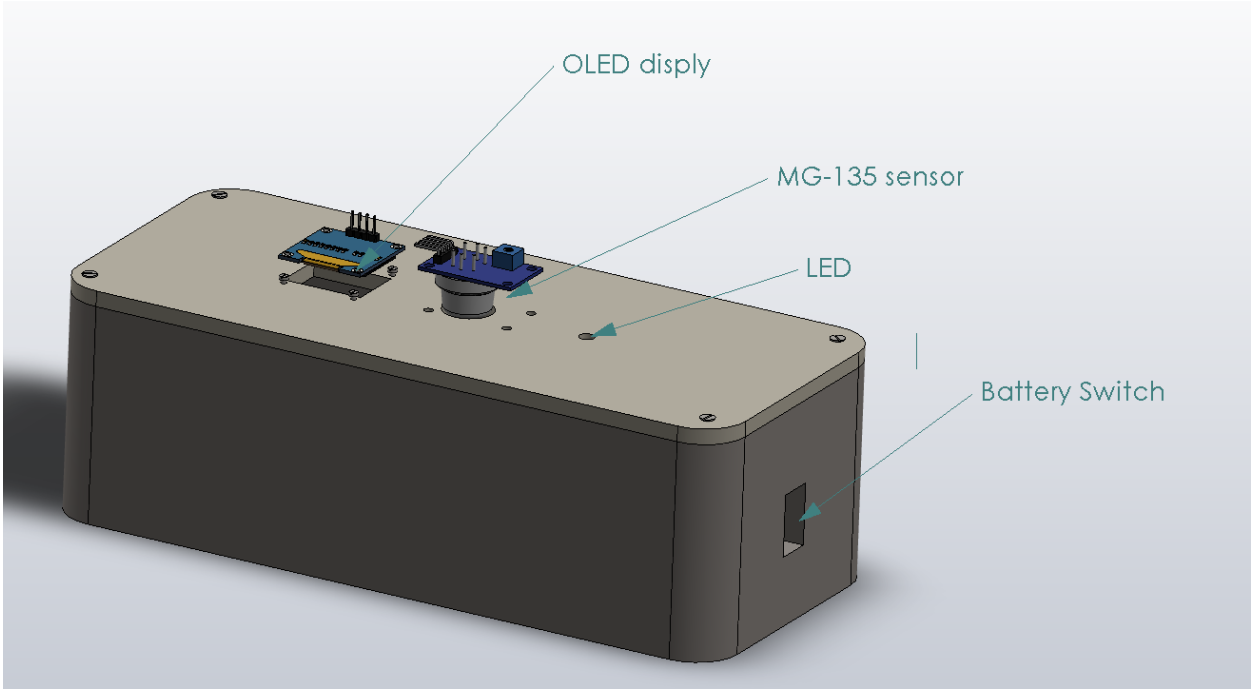


Figure 23: A 3D model of the case.

3.2 Important findings

3.2.1 Threshold Confirmation:

Through experimentation, the chosen CO2 sensor effectively detects CO2 concentration, and the system is programmed to activate an LED when CO2 levels reach 1000 ppm, providing a clear visual alert.

3.2.2 Custom enclosure design and fabrication:

A custom enclosure was designed using SolidWorks and fabricated using a 3D printer. This enclosure houses all project components, including the OLED, LED, sensor, and battery switch, demonstrating successful physical integration of the system.

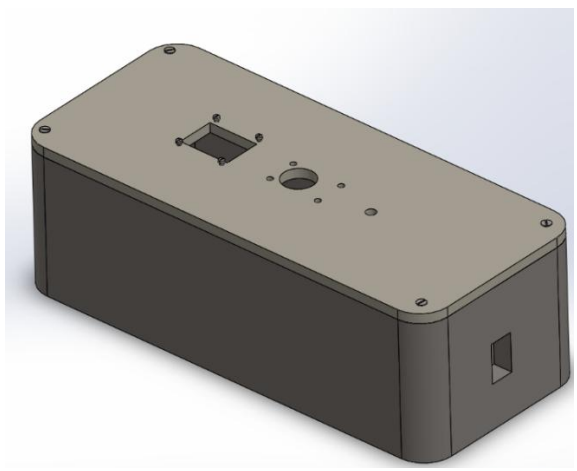


Figure 24: Enclosure manufactured using a 3D printer.

3.2.3 Practical Implications:

The prototyping stage revealed that the system was able to accurately measure and display CO2 emissions levels in real-time.

3.3 Difficulties and challenges during the testing and the prototyping stage

For sure, many problems were faced during the prototyping and testing stages of operating this project, such as programming errors and technical glitches, or even during the printing stage.

Facing problems has a significant impact on the overall prototyping and testing stage as it allows for necessary improvements and fine-tuning of the system. Additionally, continuous troubleshooting and problem-solving contribute to a more robust and efficient final product.

3.3.1 Enclosure Design

Throughout the project, a challenge arose concerning the precise cutting of apertures to fit components that were the right size. a technique were used to involve inserting the parts into a CAD model of the enclosure in order to solve this problem. The cutting procedure was then guided by the CAD model, which allowed to make sure the apertures were precisely the right size to correspond with the component dimensions.

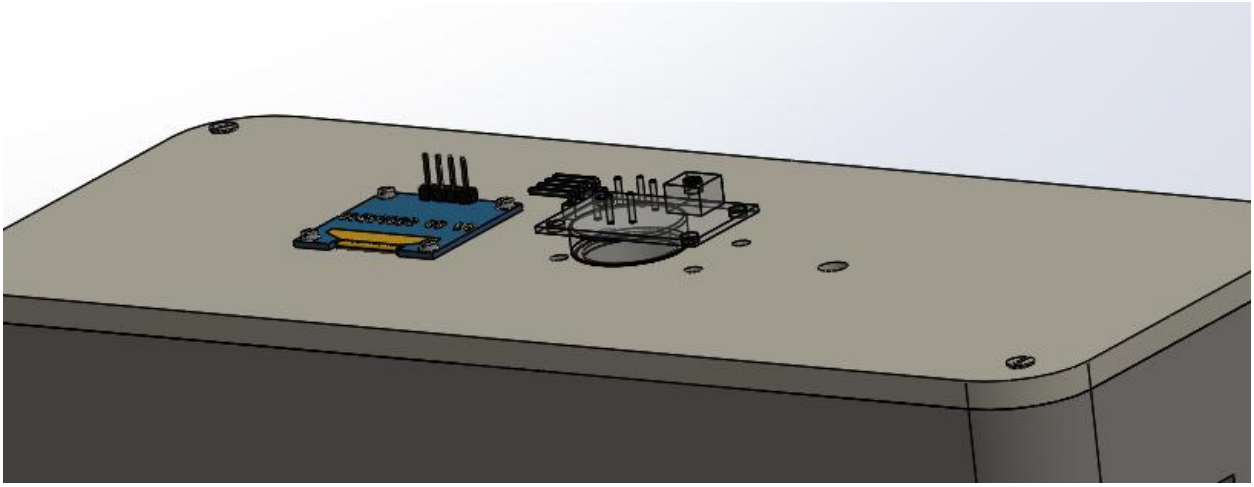


Figure 25: Entering parts into a CAD model.

3.3.2 Choosing the appropriate libraries for coding

Selecting the right libraries for coding proved to be a crucial decision in ensuring the functionality and efficiency of the project. By conducting thorough research and testing different options, I was able to identify and integrate libraries that best suited the project requirements, ultimately leading to a successful implementation.

```
#include "MQ135.h"
#include <SPI.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
```

Figure 26: Programming data and coding.

3.3.3 Choosing the appropriate sensor

Selecting the appropriate sensor is a critical step to ensure the accuracy and reliability of collected project data. This involves searching and comparing various sensor options to choose one that meets the project's specific needs and provides the necessary data for analysis.

Table 4: The different between MG-135 and MG-7.[20]

Feature	**MG-7**	**MG-135**
Sensing range	0-5000 ppm	0-10000 ppm
Sensing element	NDIR (Non-Dispersive Infrared)	NDIR (Non-Dispersive Infrared)
Response time	<10 seconds	<10 seconds
Accuracy	±30 ppm or ±3% of reading (whichever is greater)	±50 ppm or ±5% of reading (whichever is greater)
Operating temperature	-10°C to 50°C	-10°C to 50°C
Output	Analog (0-2.0V)	Analog (0-2.0V)
Power supply	3.3-5.0V DC	3.3-5.0V DC
Dimensions	31mm x 19mm x 10mm	32mm x 22mm x 14mm

Feature	MG-7	MG-135
Applications	Indoor air quality monitoring, HVAC systems.	Indoor air quality monitoring, HVAC systems, automotive emissions monitoring.

3.4 Final project

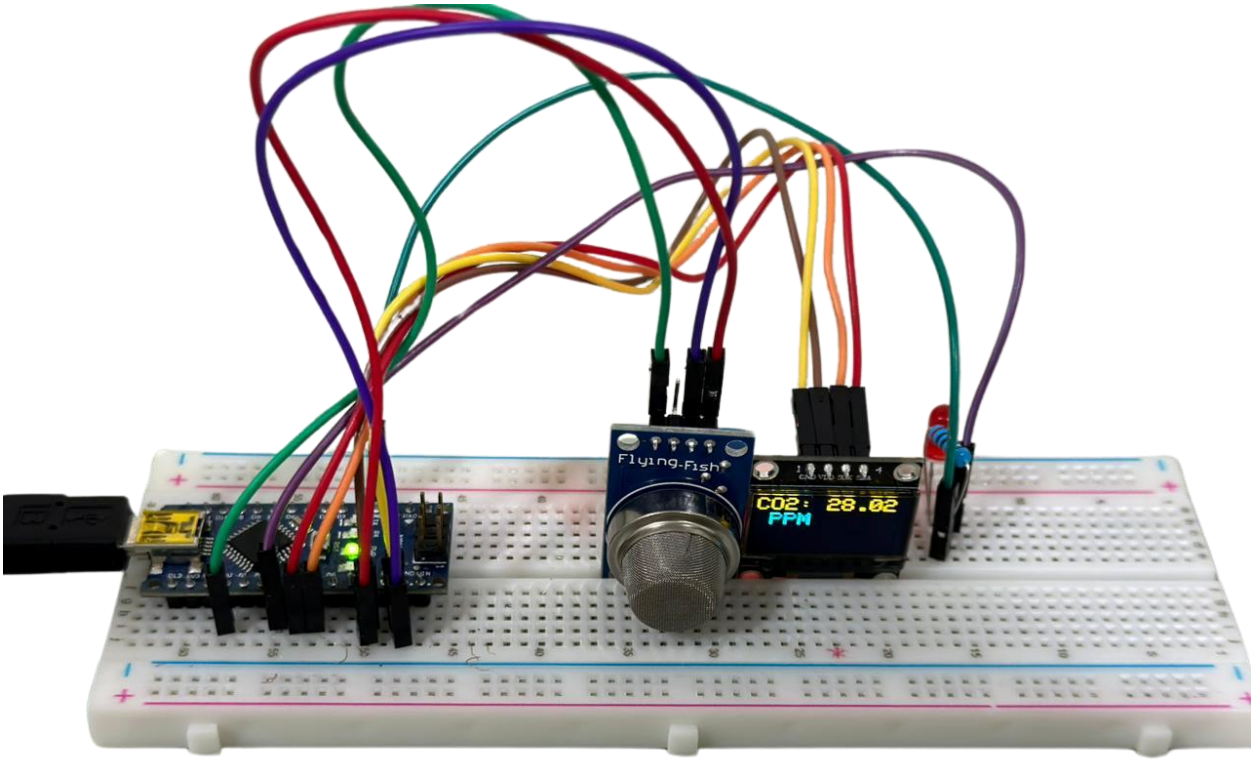


Figure 27: the final circuit of the project.

The figure above illustrates the final implementation of our project aimed at measuring CO2 emissions using Arduino. As depicted, the setup includes the Arduino board connected to the CO2 sensor, with an LED indicator integrated into the circuit. Upon reaching a threshold of 1000 parts per million (ppm) of CO2 concentration, the LED illuminates, providing a real-time visual indication of elevated CO2 levels. This effective solution demonstrates the practical application

of Arduino in environmental monitoring and showcases our commitment to addressing pressing issues such as air quality.

3.5 Final Assembled Project

Figure 28: final assembled project with LED on.

Figure 29: Final assembled project LED off.

Figures above is the fully assembled CO2 Emissions Measurement detector encased for practical use. The MG-7 sensor, Arduino board, and OLED display are strategically positioned for optimal performance and ease of access. The LED indicator is prominently placed to alert when CO2 levels surpass the 1000 ppt threshold.

4 Chapter 4. Conclusion

The conclusion of the project highlights the successful implementation of the circuit and its applications in measuring CO2 emissions. It also discusses potential future developments and improvements for the project.

4.1 The main idea of the project

The aim of the project was to implement a detector that measures CO2 emissions and to provide real-time monitoring of the data. The successful completion of the final assembly project demonstrates the effectiveness of the detector in accurately measuring CO2 emissions.

4.2 Summary of overall results

The project successfully implemented a comprehensive detector with a signal that acted when the CO2 emissions exceeded 1000 ppt. The project was developed using Arduino Nano, MG-135, OLED, and a LED.

4.3 The main findings of the project

- **Practical Implications:** The prototyping stage revealed that the system was able to accurately measure and display CO2 emissions levels in real-time.
- **Threshold Confirmation:** Through experimentation, the chosen CO2 sensor effectively detects CO2 concentration, and the system is programmed to activate an LED when CO2 levels reach 1000 ppm, providing a clear visual alert.

4.4 Recommendations for Future Work

Future devices can be developed to measure CO2 emissions in the atmosphere. There are some technical innovations that can be used for this:

1. **Use of advanced sensors:** More sensitive and accurate sensors can be developed to measure the percentage of carbon dioxide in the air. These devices may involve the use of technologies such as enhanced optical sensing or advanced chemical sensing to achieve accurate and reliable measurement.

2. **Artificial intelligence and machine learning techniques**: Artificial intelligence and machine learning techniques can be used to improve the ability of devices to analyze and understand data related to the percentage of carbon dioxide in the room. Computer models can learn from available data and improve their predictions and accuracy over time.

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